

## Comprehensive Exam (Closed Book) - PART-A

Time-: 14:00 – 17:00

Max Marks-: 30

Date: 09-05-2023

ID. No:

Name:

**Instructions:** - There are fifteen questions. Read the questions carefully and write the answers in the space provided. Every correct answer carries 2 marks while every incorrect answer carries -0.5 mark. Answer all the questions. Overwriting or ambiguity will not be considered for evaluation.

## Answers

| Q1       | Q2       | Q3       | Q4       | Q5       | Q6       | Q7       | Q8       | Q9       | Q10      | Q11      | Q12      | Q13      | Q14      | Q15      |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| <b>C</b> | <b>A</b> | <b>D</b> | <b>A</b> | <b>C</b> | <b>C</b> | <b>C</b> | <b>D</b> | <b>B</b> | <b>X</b> | <b>C</b> | <b>A</b> | <b>X</b> | <b>C</b> | <b>B</b> |

- Find the current ( $i_D$ ) through the ideal diode (zero cut in voltage and zero forward resistance) shown in **Fig.1**.  
[A] 0 A [B] 4 A [C] 1 A [D] 2 A
- Assuming opamps to be ideal, find the output voltage  $V_o$  of the circuit shown in **Fig.2**.  
[A]  $10 \cos(100t)$  [B]  $10 \sin(100t)$  [C] 0 [D]  $100 \cos(100t)$
- In the circuit shown in **Fig.3**, the op-amp is ideal, the transistor has  $V_{BE} = 0.6$  V and  $\beta = 150$ . Determine the voltage (V) at the output of the op-amp.  
[A] 10 V [B] 0 V [C] 5 V [D] 2 V
- In the programmable gain amplifier (PGA) shown in **Fig.4**, which of the following gain is not possible.  
[A] 2 [B] 4 [C] 7 [D] 13
- Find the output voltage in terms of input voltage  $V_{in}$  for the circuit shown in **Fig.5**.  
[A]  $(0.02V_{in})^{1/2}$  [B]  $(50 V_{in})^{1/2}$  [C]  $(500 V_{in})^{1/4}$  [D]  $(50V_{in})^{1/4}$
- The opamp in the circuit shown in **Fig.6** are ideal. The input signals are  $V_1 = 3 + 0.1 \sin(300 t)$  V,  $V_2 = -2 + 0.2 \sin(300 t)$  V and  $V_3 = 1$  V. Given  $R = 1$  k $\Omega$ . The time average of the output voltage  $V_o$  is \_\_\_\_\_.  
[A] 0 V [B] 0.67 V [C] 1 V [D] 1.15 V
- Calculate the total power consumed by a 8-bit flash type ADC if each comparator consumes 5 mW of power. Assume the comparators are most dominant power consuming circuits in the ADC and others are negligible.  
[A] 0.035 W [B] 0.04 W [C] 1.27 W [D] 2.56 W

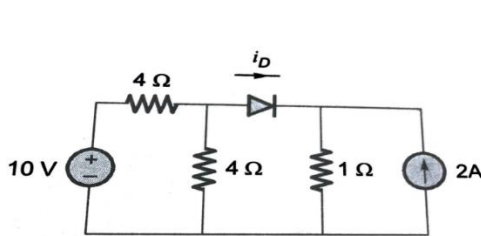


Fig.1 (Qn-1)

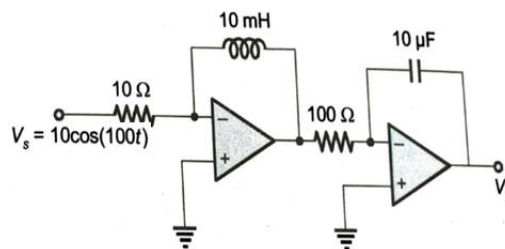


Fig.2 (Qn-2)

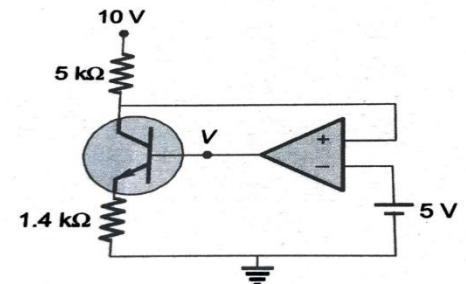


Fig.3 (Qn-3)

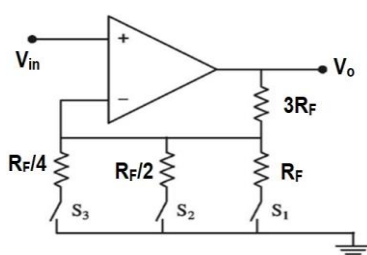


Fig.4 (Qn-4)

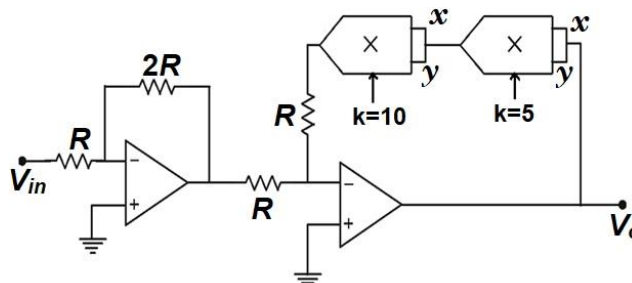


Fig.5 (Qn-5)

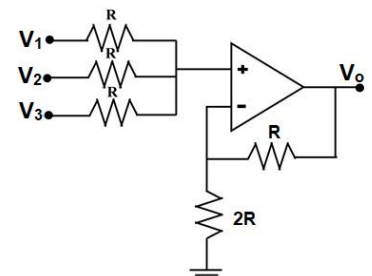


Fig.6 (Qn-6)

8. A 741 opamp is used as an inverting amplifier with a gain of 10. The voltage gain is flat upto 20 kHz. What is the maximum peak to peak signal that can be applied without distortion? Use the open loop gain of opamp to be  $10^6$  and slew rate to be  $0.5 \text{ V}/\mu\text{s}$ .
- [A] 160 mV [B] 320 mV [C] 400 mV [D] 800 mV
9. A 4-bit ladder type R-2R DAC has resolution of 0.5 V. If the reference voltage  $V_R=10\text{V}$  and the feedback resistor  $R_F$  is  $10 \text{ k}\Omega$ , the value of resistor R is
- [A]  $10 \text{ k}\Omega$  [B]  $12.5 \text{ k}\Omega$  [C]  $20 \text{ k}\Omega$  [D]  $25 \text{ k}\Omega$
10. For a given sinusoidal input, voltage waveform at the output of the clamper circuit shown in Fig.7 will be

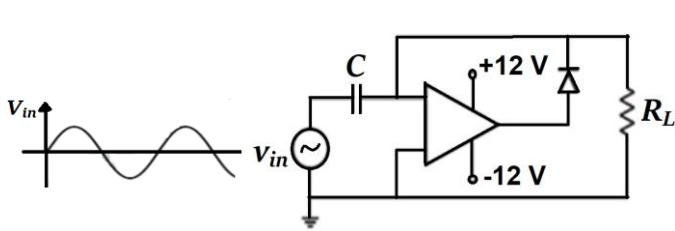
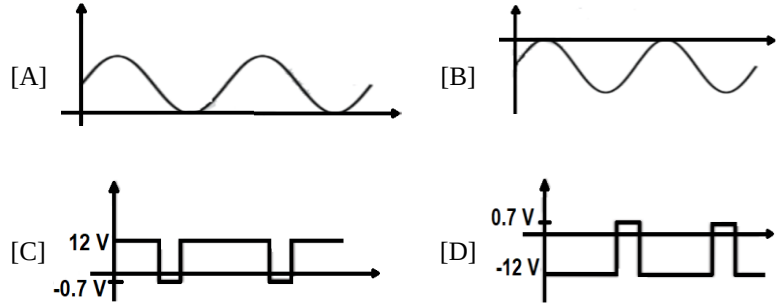


Fig.7 (Qn-10)



11. The first order all-pass filter has transfer function  $H(s) = (1-a_i*s)/(1+a_i*s)$ . If the corner frequency of this filter is 1 kHz and it has a maximum group delay of  $60 \mu\text{s}$ , the filter coefficient  $a_i$  is \_\_\_\_\_
- [A] 0.12 [B] 0.15 [C] 0.18 [D] 0.24
12. The feedback factor ( $\beta$ ) and  $R_2$  for the oscillation to occur in the circuit shown in Fig.8 when  $R_1 = 5 \text{ k}\Omega$  are
- [A]  $2/3$  and  $2.5 \text{ k}\Omega$  [B]  $1/3$  and  $10 \text{ k}\Omega$  [C]  $1/2$  and  $5 \text{ k}\Omega$  [D]  $1/4$  and  $15 \text{ k}\Omega$
13. In the triangular waveform generator shown in Fig.9,  $R_2 = 2R_1$ ,  $R = 1 \text{ k}\Omega$ ,  $R_4 = 100 \text{ k}\Omega$ ,  $C = C_1 = 0.1 \mu\text{F}$ . The value of  $R_3$  and frequency of the wave form will be \_\_\_\_\_ and \_\_\_\_\_.
- [A]  $1 \text{ k}\Omega$ ,  $2.5 \text{ KHz}$  [B]  $2 \text{ k}\Omega$ ,  $5 \text{ KHz}$  [C]  $3.14 \text{ k}\Omega$ ,  $5 \text{ KHz}$  [D]  $6.28 \text{ k}\Omega$ ,  $2.5 \text{ KHz}$
14. For the Schmitt trigger oscillator shown in the given Fig.10, the saturation output voltages are  $+10 \text{ V}$  and  $-5 \text{ V}$ . The frequency of oscillation is \_\_\_\_\_
- [A]  $2183 \text{ Hz}$  [B]  $1369 \text{ Hz}$  [C]  $869 \text{ Hz}$  [D]  $1443 \text{ Hz}$
15. Fig.11 shows a 555 timer in a monostable operation. Assuming diode to be ideal, what is the pulse width of the output pulse given  $R_i = 2 \text{ k}\Omega$ ,  $C_i = 0.01 \mu\text{F}$ ,  $R = 10 \text{ k}\Omega$ ,  $C = 0.22 \mu\text{F}$ .
- [A]  $1.2 \text{ ms}$  [B]  $2.4 \text{ ms}$  [C]  $3.6 \text{ ms}$  [D]  $4.8 \text{ ms}$

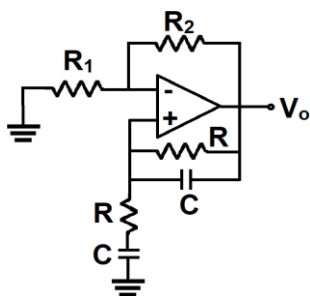


Fig.8 (Qn-12)

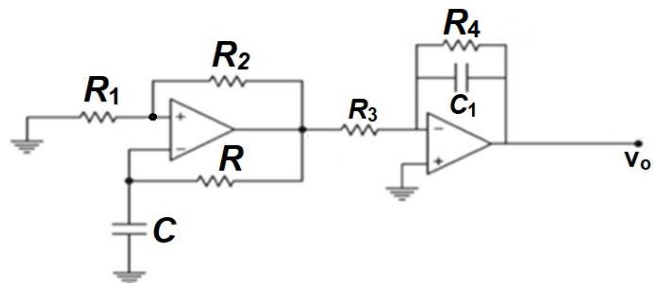


Fig.9 (Qn-13)

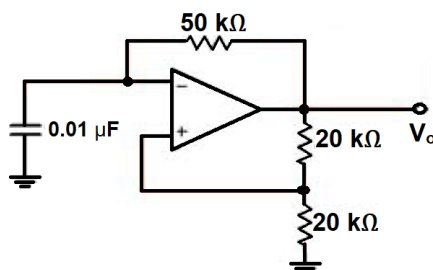


Fig.10 (Qn-14)

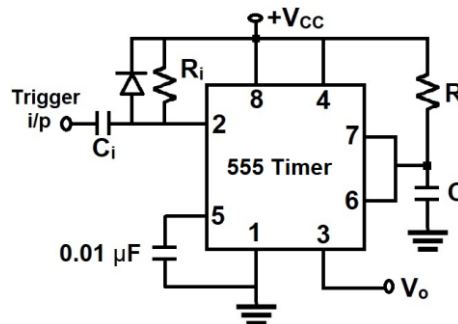


Fig.11 (Qn-15)

- 1 – Ground
- 2 – Trigger
- 3 – Output
- 4 – Reset
- 5 – Control Voltage
- 6 – Threshold
- 7 – Discharge
- 8 –  $V_{CC}$